

Auditory Kernels - Learning acoustic building blocks from speech

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Abstract

- Unlike linguistic or lexical units, such as phonemes or syllables, there are no obvious information-carrying acoustic units directly observable in the speech waveform.
- The efficient coding hypothesis (Barlow, 1961) posits that the auditory system compresses sensory information.
- Smith and Lewicki (2006) demonstrated that, when learning sparse acoustic building blocks—referred to here as *auditory kernels*—from English speech, the resulting kernels closely resemble responses measured in cat auditory systems.
- We extend their analysis to different types of speech.
- The kernels learned on the languages have a similar spectral centroid-spread relationship, and we expect that the kernels can be linked to phonetic properties.

Method

- The speech signal $s(t)$ is represented as a sum of N scaled (α_n) and shifted (τ_n) auditory kernels ϕ_n (Fig. 1):

$$s(t) = \sum_{n=0}^{N-1} \alpha_n \phi_n(t - \tau_n).$$

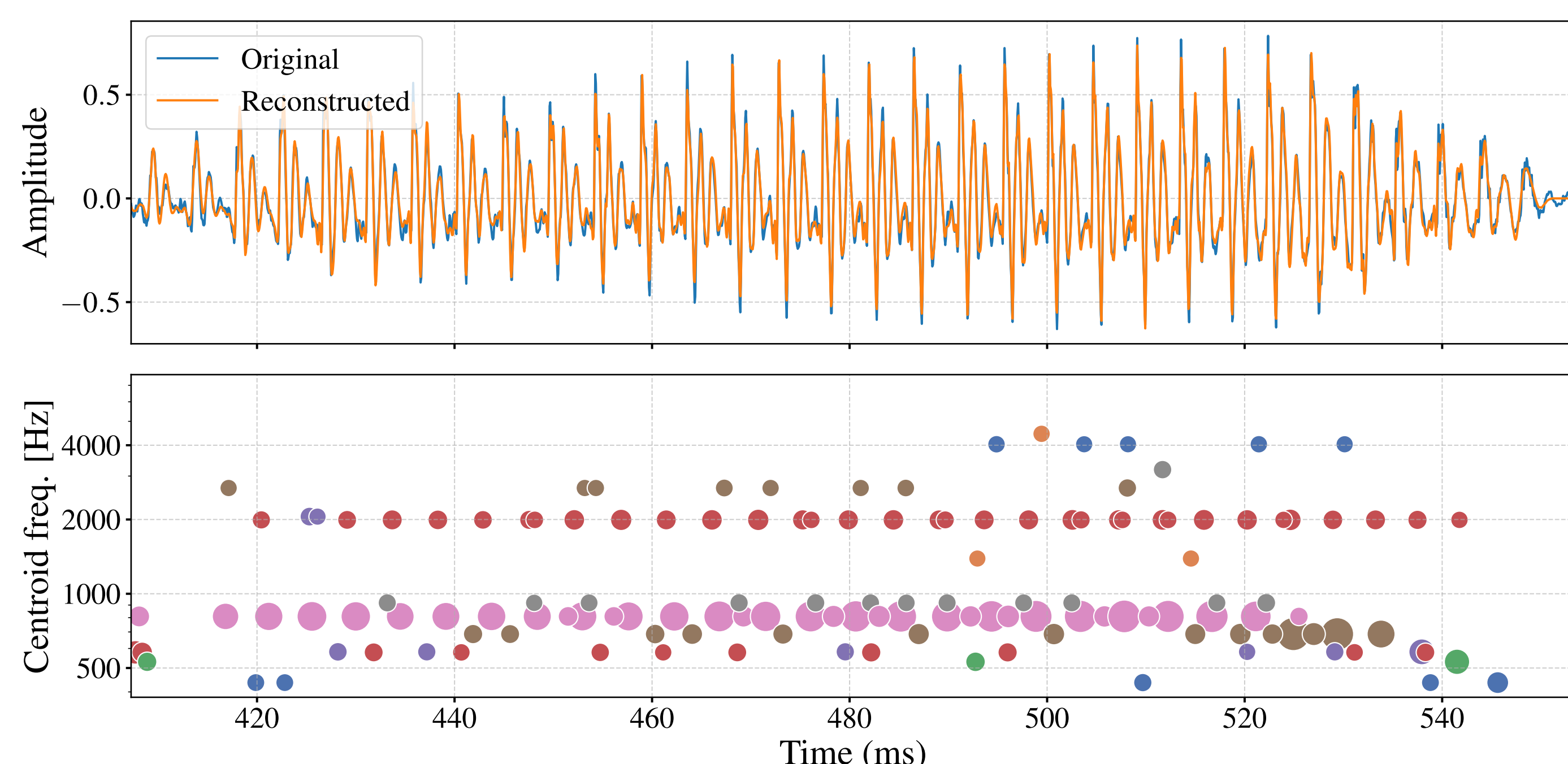


Figure 1: Speech segment of female speaker pronouncing /ae/ (as in “had”), with reconstruction and kernel activations. Point size represents $|\alpha_n|$, x-axis position τ_n , and y-axis the centroid frequency of kernel ϕ_n .

- We learn 32 unique kernels per language. Each of them can be used multiple times at different scales and delays.
- The kernels are analysed based on their spectral centroid frequency f_c and spectral spread.
- Following Conneau et al. (2017; Fleurs), the 102 languages (17 families) are divided in 7 geographic regions: Western Europe (WE, 25), Sub-Saharan Africa (SSA, 20), Eastern Europe (EE, 16), South Asia (SA, 14), Central-Asia/Middle-East/North-Africa (CMN, 12), South-East Asia (SEA, 11) and Chinese, Japanese, Korean (CJK, 4).
- We also had a look at dysarthric speech (Kim et al., 2008; UASpeech) and voiced/unvoiced speech (Garofolo et al., 1993; TIMIT).

Results

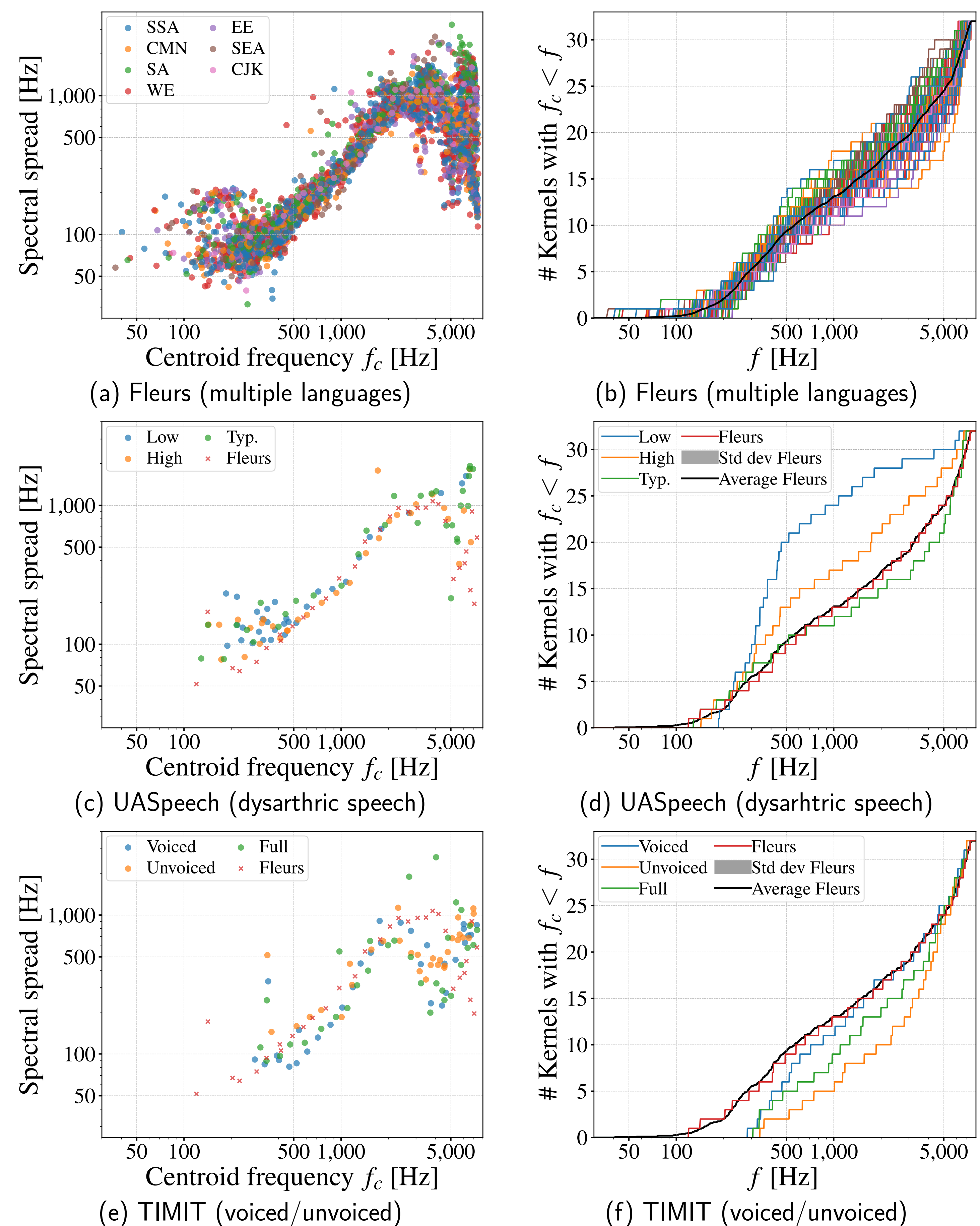


Figure 2: Spectral properties of the learned kernels. **Left column:** the centroid frequency - spectral spread relationship. **Right column:** the number of kernels with a centroid frequency below f , mean and std dev are shown. **First row:** the 102×32 kernels learned on the different languages of Fleurs. **Second row:** The kernels learned on dysarthric speakers from UASpeech (divided in high intelligibility, low intelligibility, and typical speakers). **Third row:** the kernels learned on the full, voiced, and unvoiced segments of TIMIT.

Conclusion

- We investigated spectral properties of kernels learned to efficiently represent the speech waveform.
- The consistency of spectral structure across languages suggests that the kernels capture a low-level representation which can be linked to both speech production and reception features.
- For low-intelligibility dysarthric speakers, most kernels are used to encode low-centroid frequency components.
- For unvoiced sounds, more kernels are used to encode higher centroid frequency components.
- As future work, we will investigate if the kernels can be related to phonetic properties.